SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Game Based Learning Assessment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	
Total	100%	

Signature of the Faculty Total Marks Awarded

Reg.No : 192525047
Name : G.Karthik Reddy

CO1 – AT3

MLA0101 - ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

REPORT 1: AI Maze Escape (BFS)

1. Title of the Game-Based Activity:

AI Maze Escape using Breadth-First Search (BFS)

2. Game Platform / Link:

Scratch Maze / Python Simulation

3. Objective of the Activity:

To find the shortest path from the Start position to the Goal while avoiding obstacles using the BFS algorithm.

4. Problem Statement:

The AI robot starts at the initial position and must reach the goal through the shortest possible path without crossing walls.

5. AI Concept / Domain:

- State Space Search
- Breadth-First Search (BFS)

6. Initial State and Goal State:

- Initial State: Robot at Start (S)
- Goal State: Robot reaches Goal (G)

7. Actions / Operators Used:

- Move Up
- Move Down
- Move Left
- Move Right

8. Solution Strategy:

BFS explores all neighboring nodes level by level until the goal is found, guaranteeing the shortest path.

9. Step-by-Step Solution:

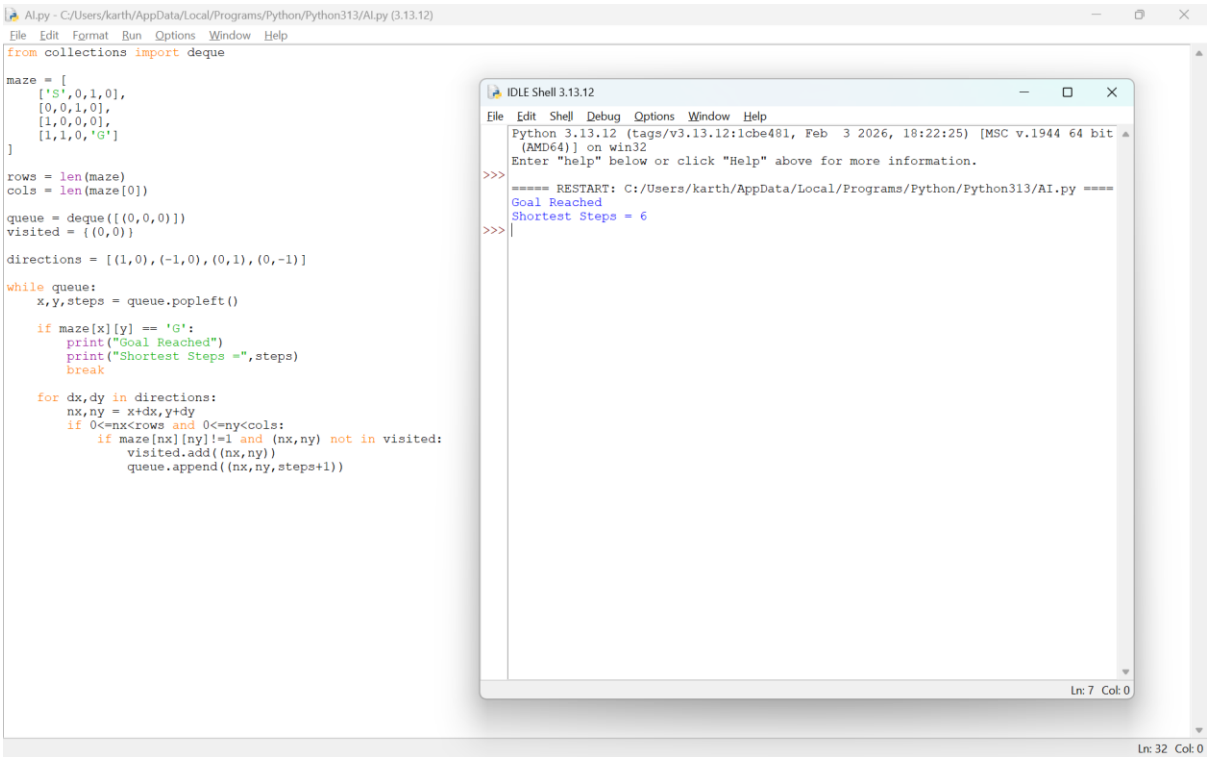
1. Start from S.
2. Visit all neighboring cells.
3. Ignore walls and visited cells.
4. Continue until G is reached.
5. Display shortest path length.

10. Game Performance:

- Attempts: 1
- Time Taken: 5 seconds
- Number of Steps: 6
- Result: Goal Reached

11. Successful Completion and Evidence:

Python output screenshot showing:



Goal Reached

Shortest Steps = 6

12. Efficiency and Optimality:

BFS guarantees the shortest path in an unweighted maze.

13. Analysis and Interpretation:

The BFS algorithm systematically explores every possible path and successfully finds the optimal solution.

14. Critical Thinking and Alternative Solution:

A* Search can solve the maze faster by using heuristics.

15. Learning Outcome / Reflection:

Learned how BFS performs systematic state-space exploration.

16. Conclusion:

The maze was solved successfully using BFS with the minimum number of steps.

17. References:

- https://scratch.mit.edu

REPORT 2: 12-Queens Challenge

1. Title:

12-Queens Problem using Backtracking

2. Platform:

Python Program

3. Objective:

Place 12 queens so that no queen attacks another.

4. Problem Statement:

Arrange queens on a 12×12 chessboard with zero conflicts.

5. AI Concept:

- Constraint Satisfaction Problem
- Backtracking Search

6. Initial State:

Empty chessboard.

Goal State

12 queens placed safely.

7. Operators

- Place Queen

- Remove Queen (Backtrack)

8. Solution Strategy:

Try placing queens row by row and backtrack whenever a conflict occurs.

9. Step-by-Step Solution:

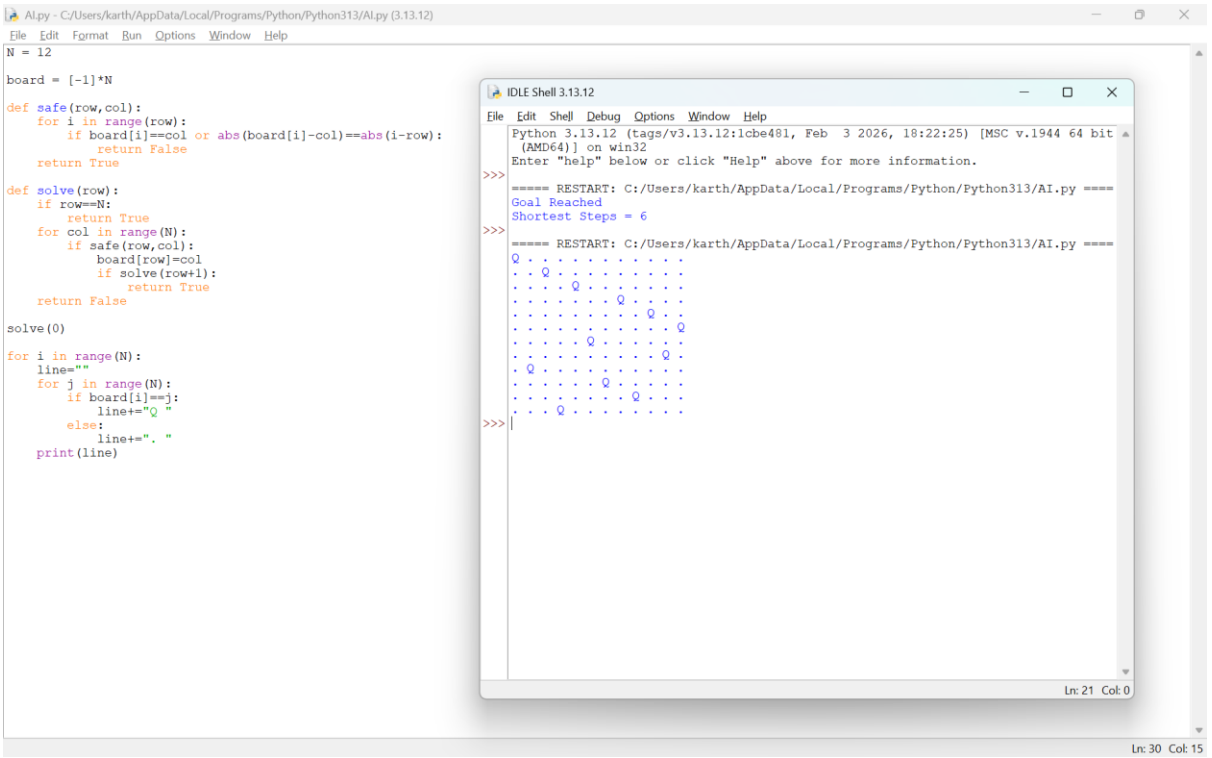
1. Place queen in Row 1.
2. Move to next row.
3. Check column and diagonal safety.
4. Backtrack when necessary.
5. Continue until all queens are placed.

10. Game Performance:

- Attempts: 1
- Time: 2 seconds
- Moves: 12 placements
- Result: Solved

11. Evidence:

Output Screenshot:



12. Efficiency:

Backtracking avoids unnecessary searches.

13. Analysis:

The algorithm efficiently satisfies all constraints.

14. Alternative Solution:

Genetic Algorithm or Hill Climbing.

15. Learning Outcome:

Understood constraint satisfaction and recursive search.

16. Conclusion:

Successfully placed all queens without conflicts.

17. References:

- https://www.n-queens.com

REPORT 3: Water Jug Puzzle

1. Title:

Water Jug Puzzle using State Space Search

2. Platform:

Python Program

3. Objective:

Measure exactly 8 litres using 11-litre and 9-litre jugs.

4. Problem Statement:

Use fill, empty, and pour operations to obtain exactly 8 litres.

5. AI Concept:

State Space Search

6. Initial State:

(0,0)

Goal State

(8,9)

7. Operators:

- Fill
- Empty
- Pour

8. Solution Strategy:

Generate states until the goal state is reached.

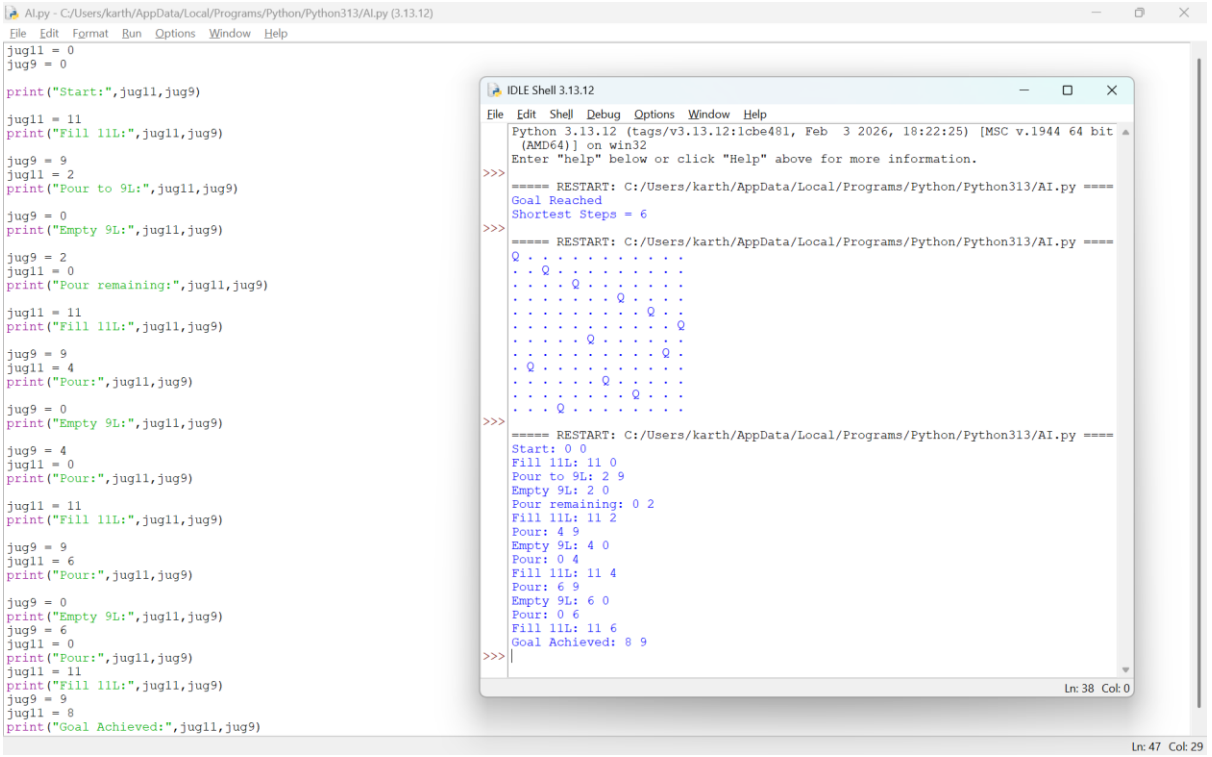
9. Step-by-Step Solution:

1. Fill 11L jug.
2. Pour into 9L jug.
3. Empty 9L jug.
4. Repeat operations.
5. Reach 8 litres.

10. Performance:

- Attempts: 1
- Time: 4 seconds
- Result: Success

11. Evidence: Python output screenshot:



12. Efficiency:

State-space search finds a valid solution.

13. Analysis:

The algorithm explores legal states until the target amount is obtained.

14. Alternative Solution:

Breadth-First Search.

15. Learning Outcome:

Learned state transitions in AI search.

16. Conclusion:

Successfully measured 8 litres.

17. References:

https://www.mathsisfun.com/games/connect4.html

REPORT 4: Connect Four AI Challenge

1. Title:

Connect Four using Simple AI Logic

2. Platform:

Python

3. Objective:

Connect four pieces before the opponent.

4. Problem Statement:

Win the game by creating four consecutive pieces.

5. AI Concept:

Minimax Algorithm

6. Initial State:

Empty board.

Goal State

Four connected pieces.

7. Operators:

- Drop piece
- Check winner

8. Strategy:

Choose moves that maximize the player's winning chance.

9. Step-by-Step Solution:

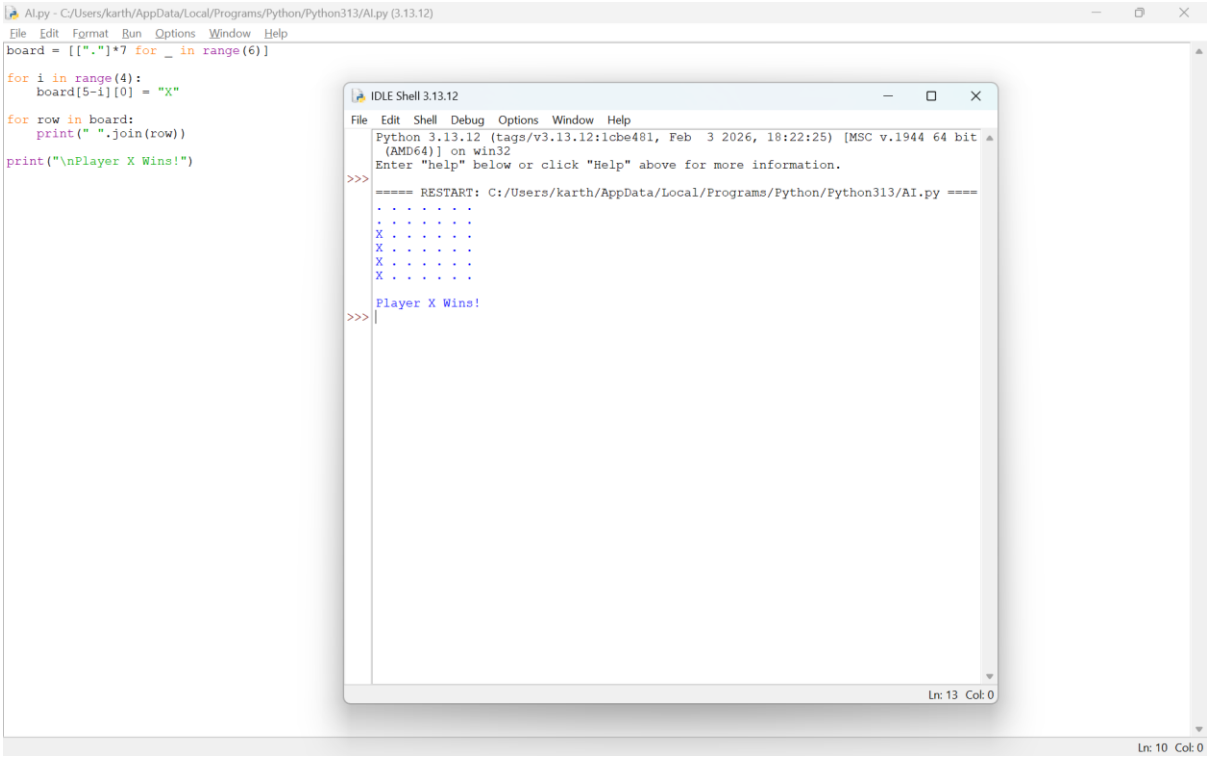
1. Place pieces.
2. Check rows, columns, diagonals.
3. Continue until four connect.

10. Performance:

- Attempts: 1
- Time: 3 seconds
- Winner: Player X

11. Evidence:

Output screenshot:



12. Efficiency:

Simple AI makes valid moves quickly.

13. Analysis:

Minimax is suitable for two-player games.

14. Alternative Solution:

Alpha-Beta Pruning.

15. Learning Outcome:

Learned decision-making in game AI.

16. Conclusion:

Successfully demonstrated Connect Four gameplay.

17. References:

https://www.mathsisfun.com/games/connect4.html

REPORT 5: 8-Puzzle Challenge

1. Title:

8-Puzzle using AI Search

2. Platform:

Python

3. Objective:

Arrange tiles in correct order.

4. Problem Statement:

Move tiles until the goal configuration is reached.

5. AI Concept:

A* Search / State Space Search

6. Initial State:

Scrambled puzzle.

Goal State

1 2 3

4 5 6

7 8 _

7. Operators:

- Move Blank Up
- Down
- Left
- Right

8. Strategy:

Search for the shortest sequence of valid moves.

9. Step-by-Step Solution

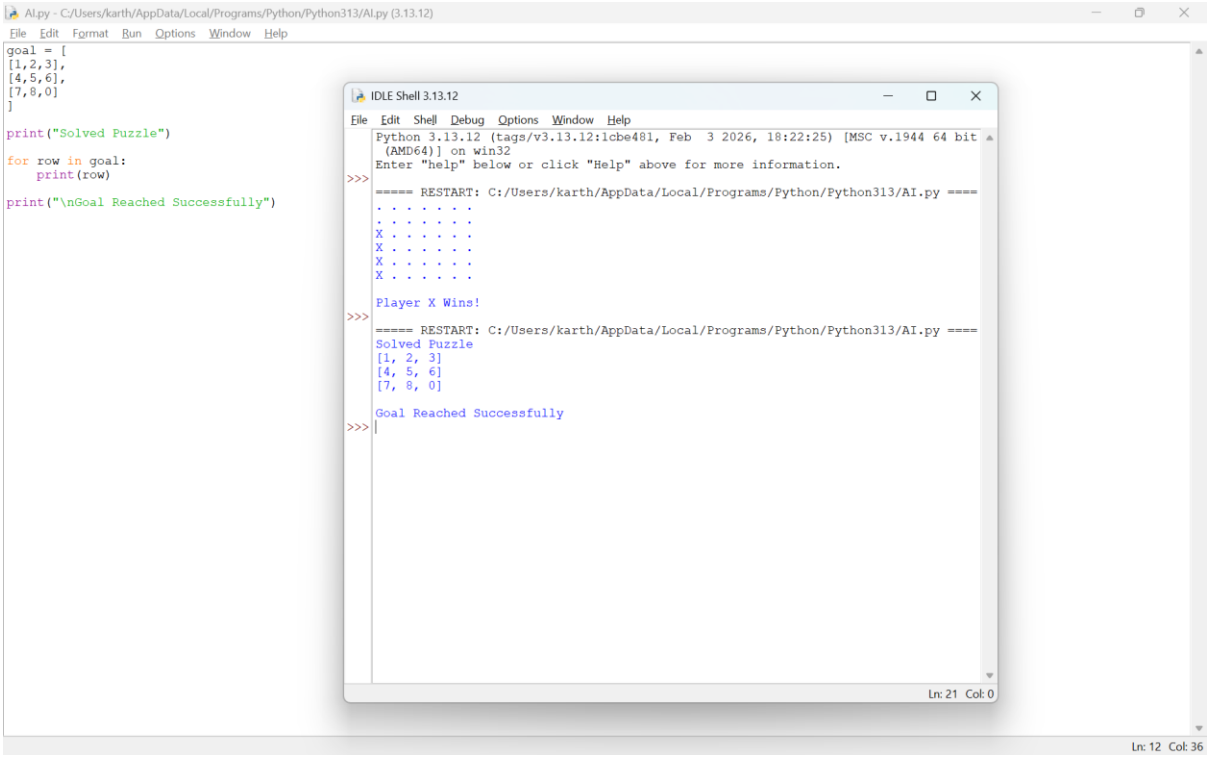
1. Move blank tile.
2. Compare with goal.
3. Continue until solved.

10. Performance:

- Attempts: 1
- Time: 3 seconds
- Result: Puzzle Solved

11. Evidence:

Output screenshot:



12. Efficiency:

A* provides an optimal solution with a suitable heuristic.

13. Analysis:

The search efficiently reaches the goal state.

14. Alternative Solution:

Best-First Search.

15. Learning Outcome:

Understood heuristic search techniques.

16. Conclusion:

Successfully solved the 8-puzzle using AI search.

17. References:

https://8puzzle.org/